📋 Solution Requirements

FlightFinder - Functional & Non-Functional Requirements

1. Functional Requirements

1.1 User Management

Req ID	Requirement	Priority	Status
FR-001	System shall allow users to register with username, email, and password	High	Required
FR-002	System shall encrypt passwords using bcrypt	High	Required
FR-003	System shall support three user types: Customer, Flight Operator, Admin	High	Required
FR-004	System shall provide instant approval for customers	High	Required
FR-005	System shall require admin approval for flight operators	High	Required
FR-006	System shall allow users to login with email and password	High	Required
FR-007	System shall maintain user sessions	High	Required

1.2 Flight Search & Display

Req ID	Requirement	Priority	Status
FR-008	System shall allow search by origin, destination, and date	High	Required
FR-009	System shall disable past dates in date picker	High	Required
FR-010	System shall display flight details (name, ID, timings, price)	High	Required
FR-011	System shall show available seats for each flight	Medium	Required
FR-012	System shall support return journey search	Medium	Optional

1.3 Booking Management

Req ID	Requirement	Priority	Status
FR-013	System shall allow booking for multiple passengers	High	Required
FR-014	System shall support four seat classes (Economy, Premium, Business, First)	High	Required
FR-015	System shall calculate price dynamically based on class and passengers	High	Required

FR-016	System shall assign seats automatically with class codes	Medium	Required
FR-017	System shall store passenger details (name, age)	High	Required
FR-018	System shall provide instant booking confirmation	High	Required
FR-019	System shall allow users to view their bookings	High	Required
FR-020	System shall allow users to cancel bookings	High	Required

1.4 Flight Operator Features

Req ID	Requirement	Priority	Status
FR-021	System shall allow operators to add new flights	High	Required
FR-022	System shall allow operators to edit flight details	High	Required
FR-023	System shall show operators only their own flights	High	Required
FR-024	System shall allow operators to view bookings for their flights	High	Required
FR-025	System shall display approval status to operators	High	Required

1.5 Admin Features

Req ID	Requirement	Priority	Status
FR-026	System shall allow admin to view all users	High	Required
FR-027	System shall allow admin to approve operator registrations	High	Required
FR-028	System shall allow admin to reject operator registrations	High	Required
FR-029	System shall allow admin to view all flights	High	Required
FR-030	System shall allow admin to view all bookings	High	Required

2. Non-Functional Requirements

2.1 Performance Requirements

- **NFR-001:** System shall load pages within 2 seconds
- **NFR-002:** System shall handle 100 concurrent users
- **NFR-003:** Database queries shall execute within 500ms
- **NFR-004:** API response time shall be under 1 second

2.2 Security Requirements

- **NFR-005:** All passwords shall be encrypted using bcrypt with salt rounds
- **NFR-006:** System shall implement role-based access control (RBAC)
- **NFR-007:** System shall protect against SQL injection attacks
- **NFR-008:** System shall use HTTPS for all communications
- **NFR-009:** System shall validate all user inputs
- **NFR-010:** System shall implement CORS protection

2.3 Usability Requirements

- **NFR-011:** Interface shall be intuitive requiring no training
- **NFR-012:** System shall provide clear error messages
- **NFR-013:** System shall be responsive on all devices
- **NFR-014:** Booking process shall complete in under 5 minutes
- **NFR-015:** System shall provide visual feedback for all actions

2.4 Reliability Requirements

- **NFR-016:** System shall have 99% uptime
- **NFR-017:** System shall handle errors gracefully
- **NFR-018:** System shall maintain data integrity
- **NFR-019:** System shall backup data daily

2.5 Scalability Requirements

- **NFR-020:** System shall support horizontal scaling
- **NFR-021:** Database shall handle 10,000+ records efficiently

- **NFR-022:** System architecture shall be modular for easy updates

2.6 Compatibility Requirements

- **NFR-023:** System shall work on Chrome, Firefox, Safari, Edge
- **NFR-024:** System shall be compatible with Windows, macOS, Linux
- **NFR-025:** System shall work on mobile devices (responsive design)

2.7 Maintainability Requirements

- **NFR-026:** Code shall follow industry best practices
- **NFR-027:** System shall have comprehensive documentation
- **NFR-028:** Code shall be modular and reusable
- **NFR-029:** System shall use version control (Git)

3. System Constraints

Constraint Type	Description
Technology Stack	Must use MERN stack (MongoDB, Express.js, React.js, Node.js)
Database	Must use MongoDB for data storage
Frontend Framework	Must use React.js with Bootstrap 5
Backend Framework	Must use Node.js with Express.js
Port Configuration	Server: Port 6001, Client: Port 3000, MongoDB: Port 27017

4. Acceptance Criteria

System shall be considered complete when:

- All high-priority functional requirements are implemented
- All security requirements are met
- System passes user acceptance testing
- Performance benchmarks are achieved
- Documentation is complete
- System is deployed and accessible
- All user roles function correctly
- Data validation works properly
- Error handling is comprehensive

5. Success Metrics

Metric	Target
User Registration Success Rate	95%+
Booking Completion Rate	90%+
Average Booking Time	< 5 minutes
System Uptime	99%+
User Satisfaction	4.5/5 stars
Error Rate	< 1%

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